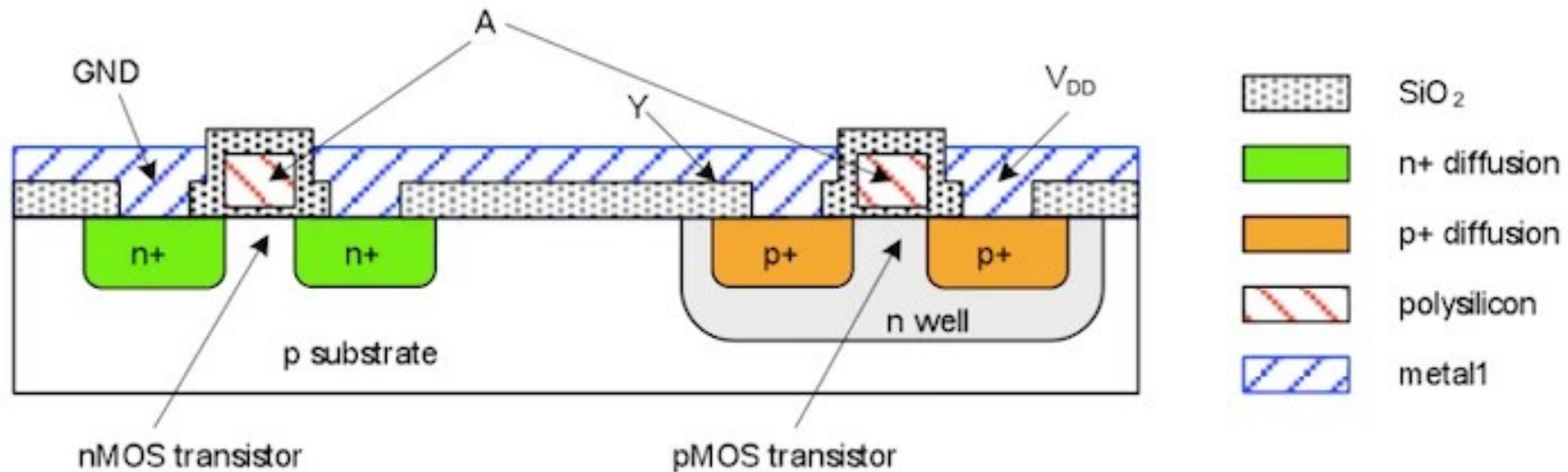


MOS Transistor Types: nMOS & pMOS

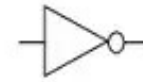
Isolation without Bulk electrode



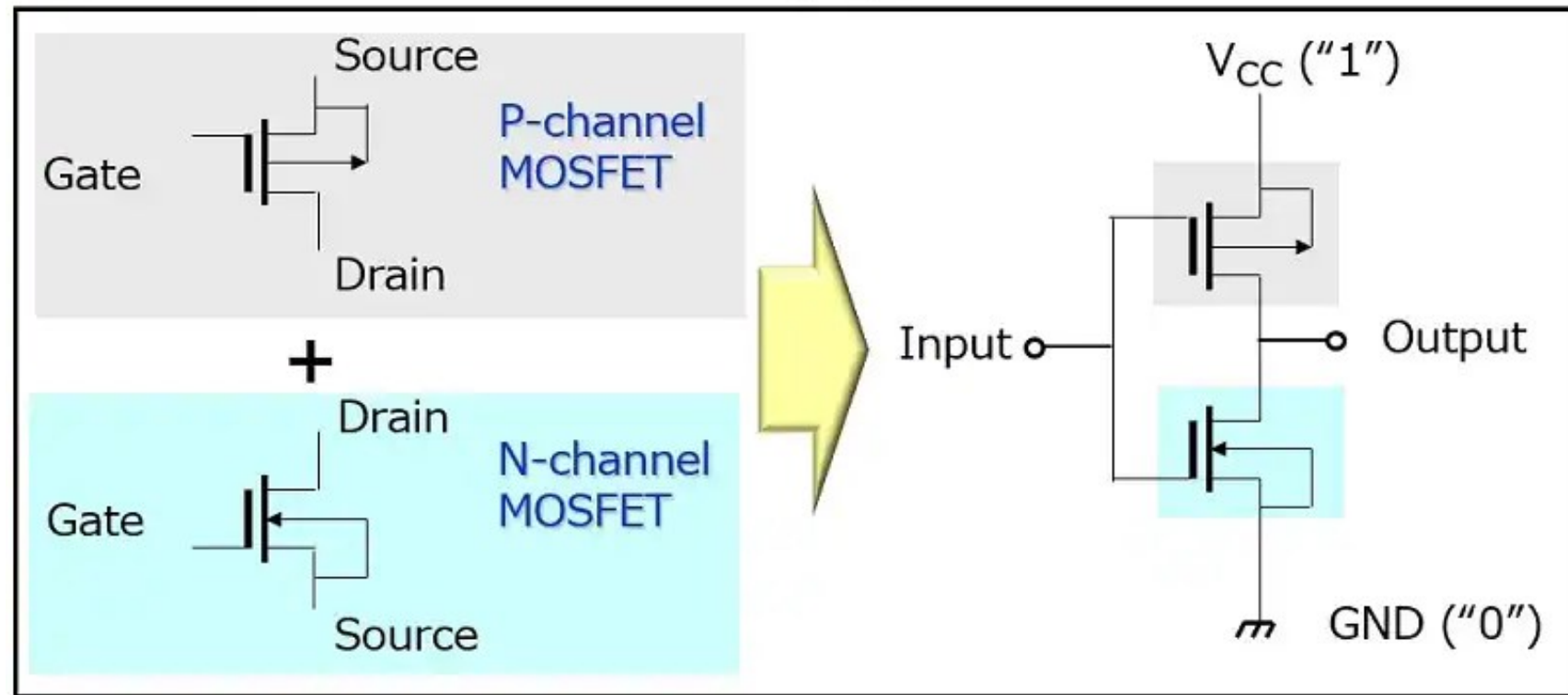
https://www.allaboutcircuits.com/uploads/articles/NMOS_and_PMOS_silicon_layouts_in_a_CMOS_technology.jpg

CMOS (Complementary Metal-Oxide-Semiconductor)

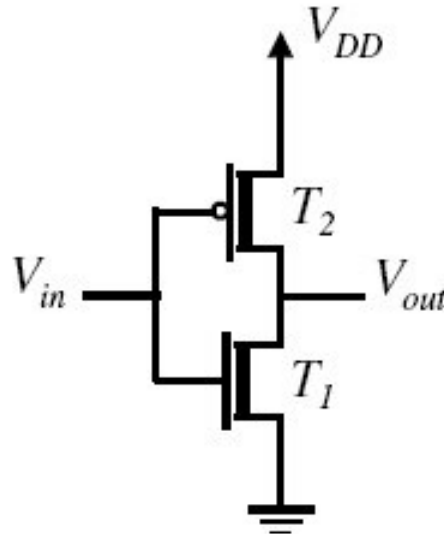
Inverter



Logic symbol



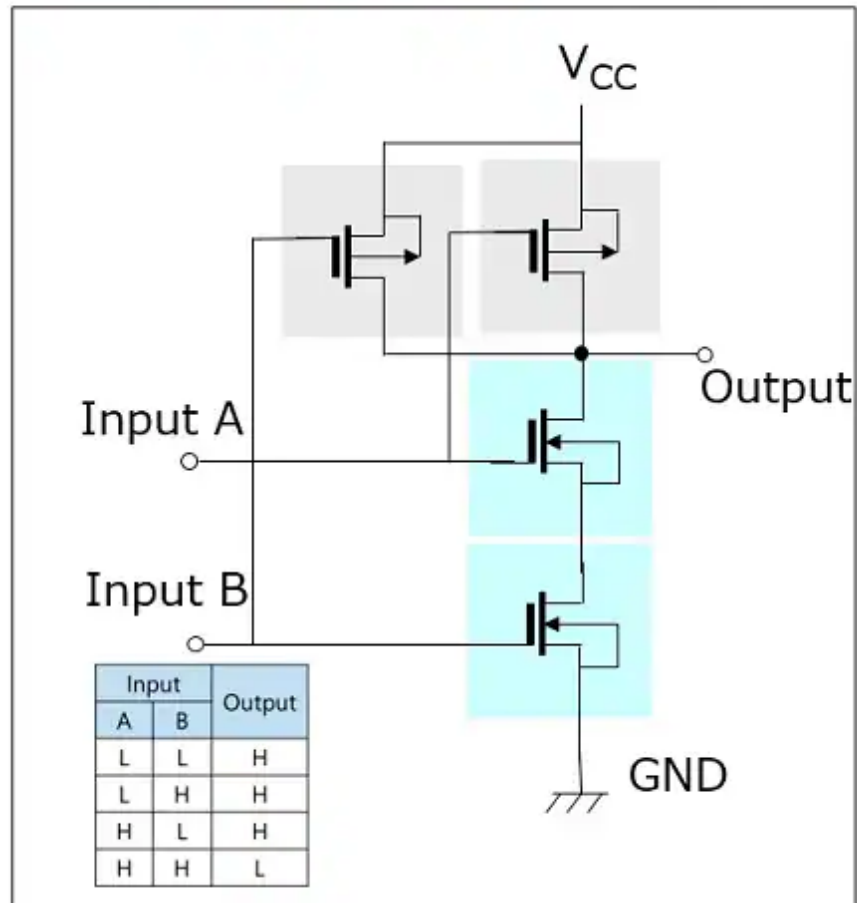
https://toshiba.semicon-storage.com/content/dam/toshiba-ss-v3/master/en/semiconductor/knowledge/e-learning/cmos-logic-basics/chap2-1_en.jpg

Truth Table	Integrated Circuit												
<table><tr><th>V_{in}</th><th>T_1</th><th>T_2</th><th>V_{out}</th></tr><tr><td>0</td><td>off</td><td>on</td><td>1</td></tr><tr><td>1</td><td>on</td><td>off</td><td>0</td></tr></table>	V_{in}	T_1	T_2	V_{out}	0	off	on	1	1	on	off	0	
V_{in}	T_1	T_2	V_{out}										
0	off	on	1										
1	on	off	0										

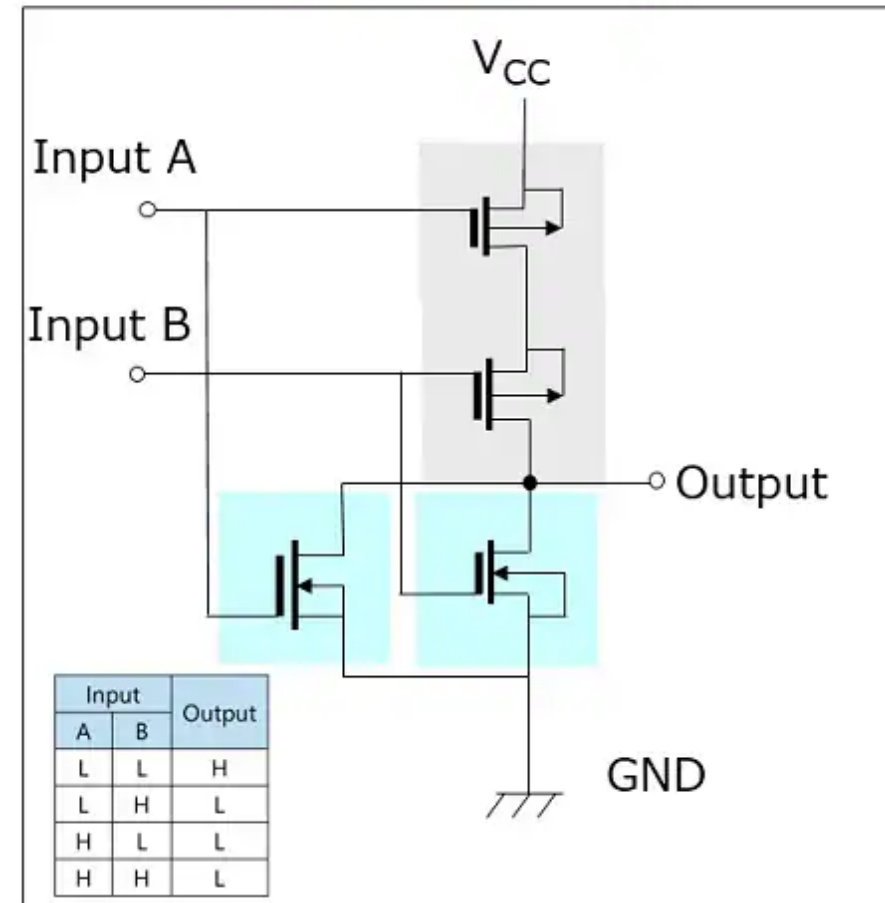
<https://www.oreilly.com/api/v2/epubs/9780470900550/files/images/ch005-f007.jpg>

CMOS NAND and NOR gates

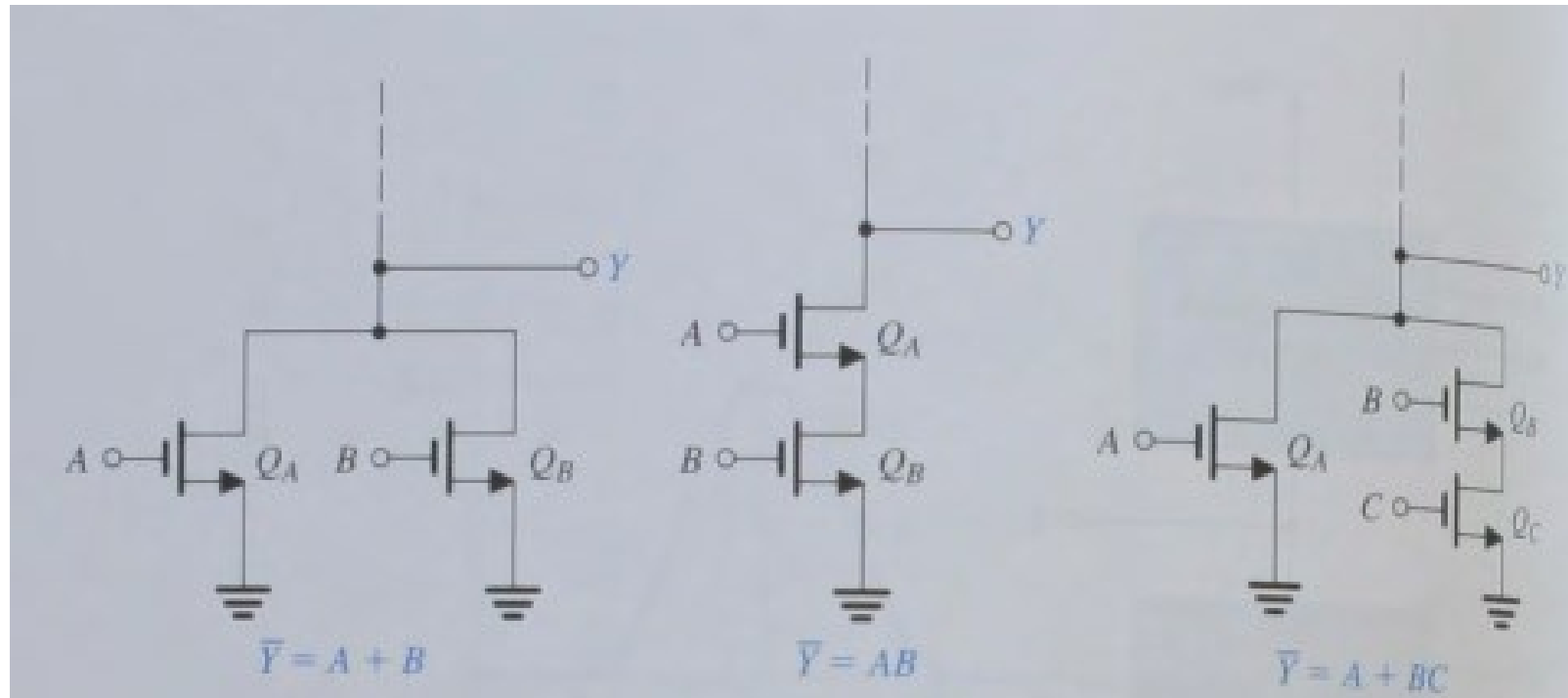
2-input NAND gate



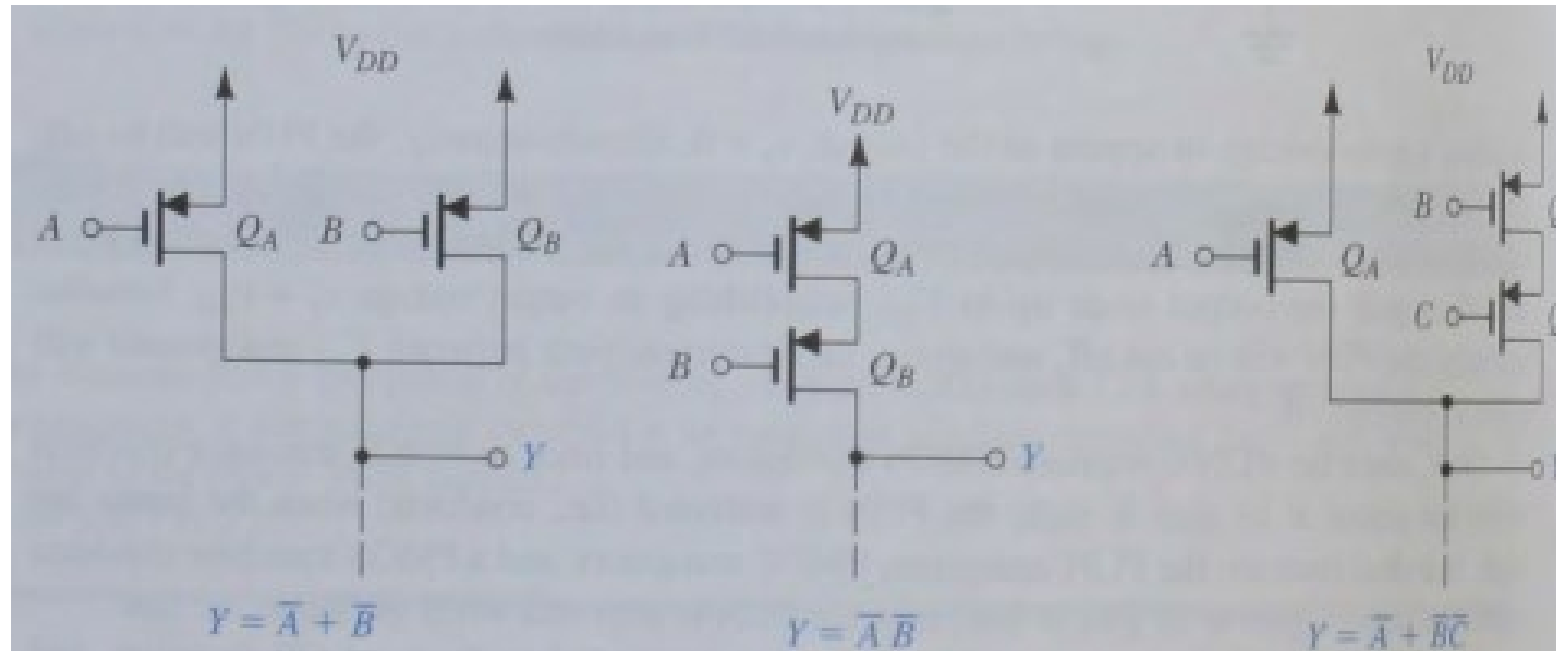
2-input NOR gate



https://toshiba.semicon-storage.com/content/dam/toshiba-ss-v3/master/en/semiconductor/knowledge/e-learning/cmos-logic-basics/chap2-2-3_en.jpg

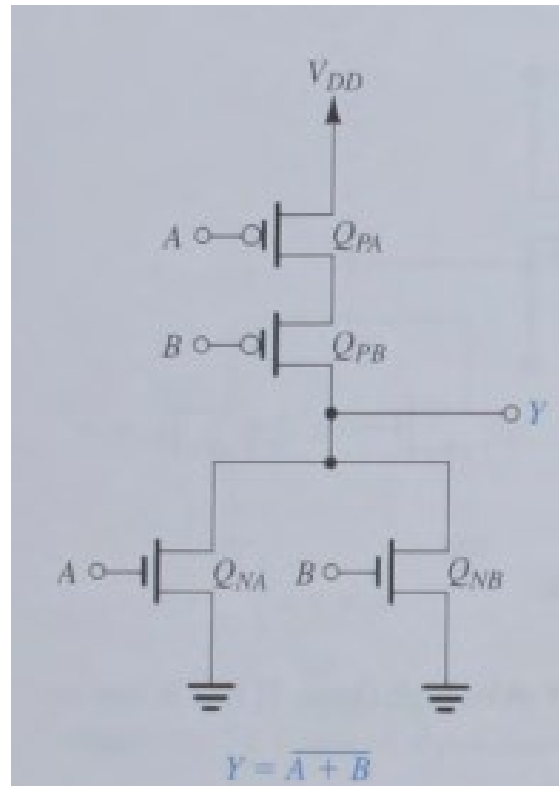


Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



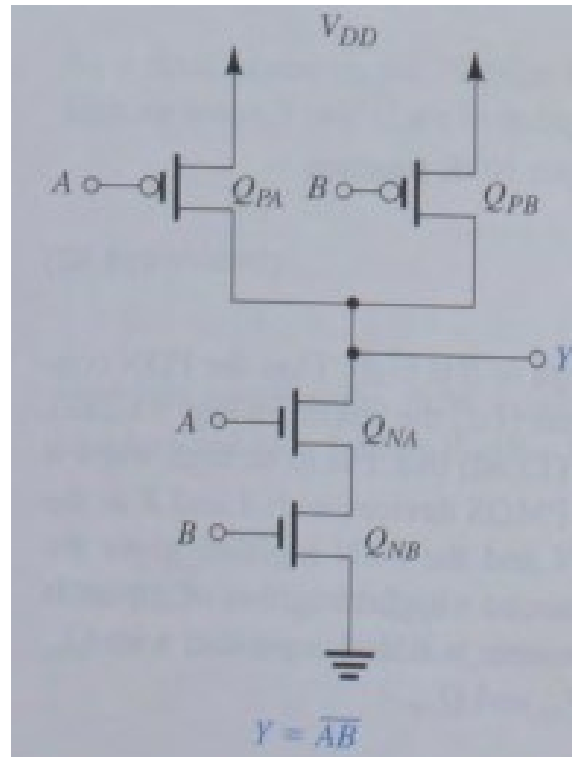
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: Two-Input NOR Gate



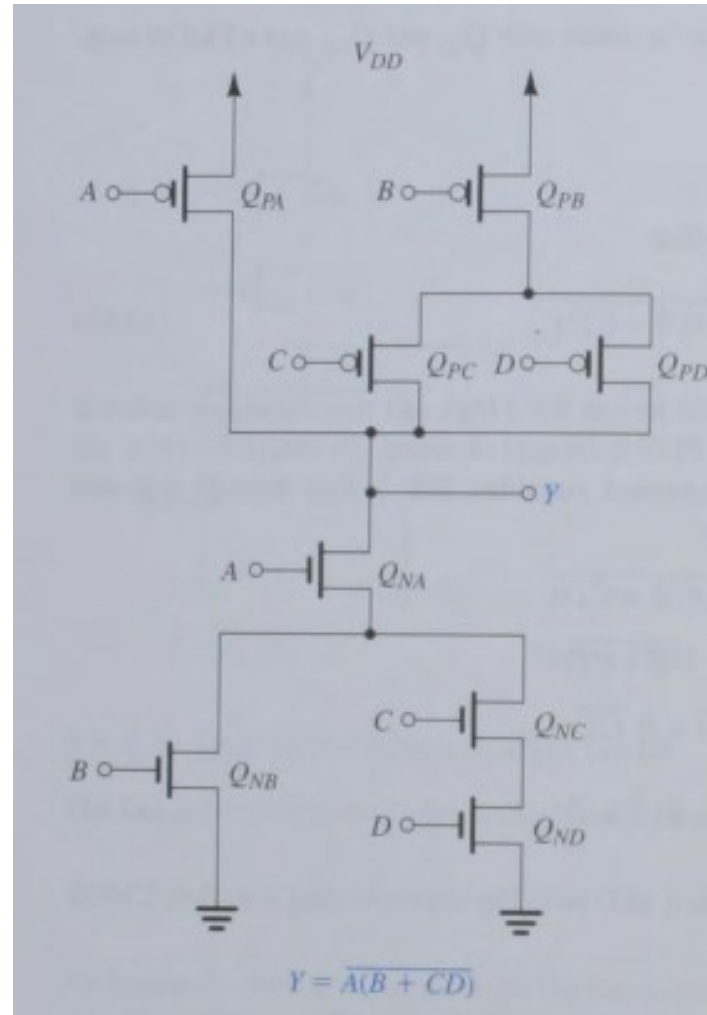
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: Two-Input NAND Gate



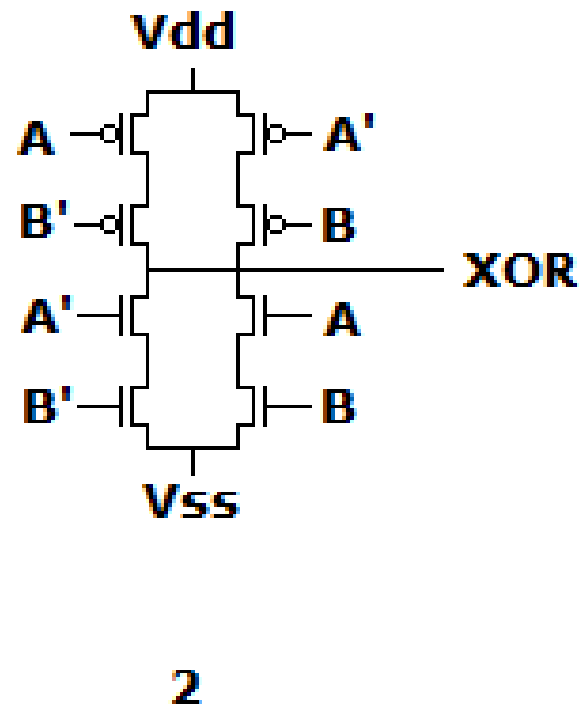
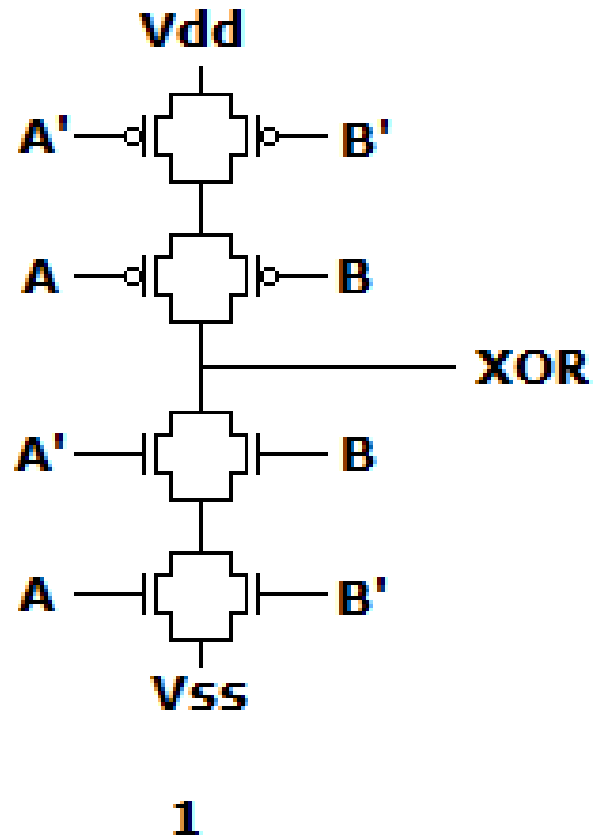
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: An Arbitrary Gate



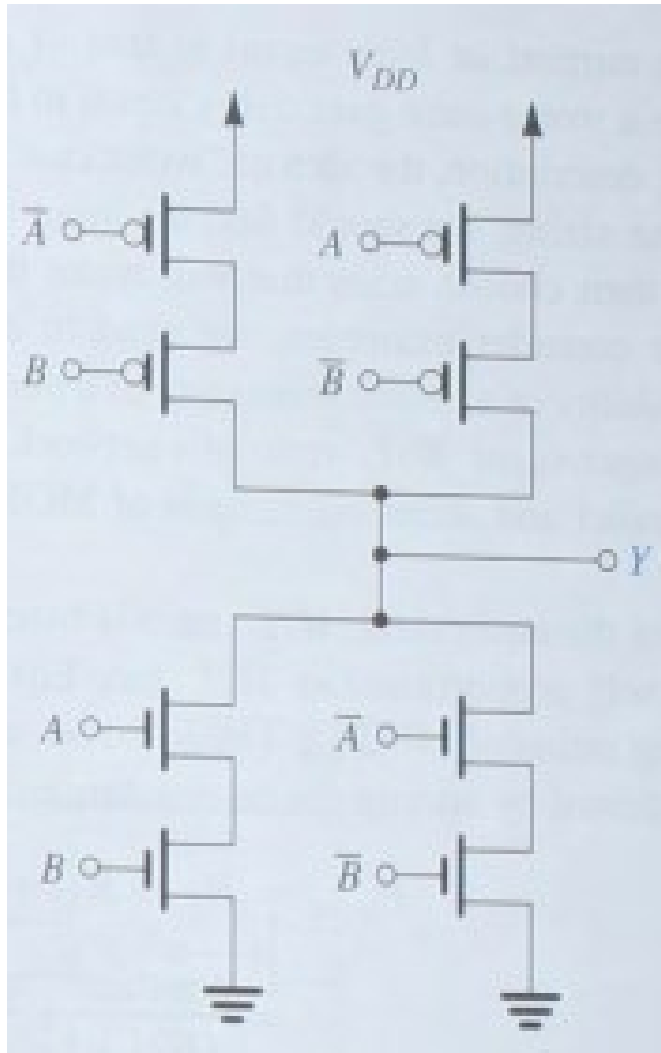
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: The XOR Gate



<https://i.sstatic.net/SACI5.png>

Applications of CMOS: The XOR Gate



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: The XOR Gate

$$\overline{Y} =$$
$$= (\overline{A\overline{B}} + \overline{\overline{A}B}) =$$
$$= (\overline{A\overline{B}})(\overline{\overline{A}B}) =$$
$$= (\overline{A+B})(\overline{A+B})$$

$$\overline{AB} = \overline{A} + \overline{B}$$
$$\overline{A+B} = \overline{A}\overline{B}$$

$$Y = A \oplus B = A\overline{B} + \overline{A}B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

The diagram illustrates the CMOS implementation of an XOR gate. It consists of two stages. The first stage is a CMOS inverter with inputs A and B, and output Y. The second stage is a CMOS NAND gate with inputs A and B, and output Y. The output of the first stage is connected to the input of the second stage. The final output is Y.

Applications of CMOS: The XOR Gate

$$\begin{aligned} \overline{Y} &= \\ &= (\overline{A\overline{B}} + \overline{\overline{A}B}) = \\ &= (\overline{A\overline{B}})(\overline{\overline{A}B}) = \\ &= (\overline{A+B})(\overline{A+B}) \end{aligned}$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$Y = A \oplus B = A\overline{B} + \overline{A}B$$

$$\begin{aligned} \overline{AB} &= \overline{A} + \overline{B} \\ \overline{A+B} &= \overline{A}\overline{B} \end{aligned}$$

Applications of CMOS: The XOR Gate

